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CRISTALLINA-Q XFEL DIFFRACTOMETERS

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Abstract

In a well-known European free-electron laser facility (SwissFEL), a new branch (ARAMIS 3) of the beamline delivers hard X-ray to the CRISTALLINA experimental hutch. CRISTALLINA-Q station inside intends to investigate advanced materials focused on specific quantum materials (QM) structures and properties. Two new dedicated Diffractometers (Dm(s)) have been developed. CrQ-$Dm_{1,2}$ are heavy load precision machines which, through adequate techniques and instruments, under extreme conditions (temp, press, rad), working in tandem are expected to advance the related investigations. CrQ-Dm_1 manipulates a large-size (h = 2.5 m) cryo-magnet (1 t, 5.2 T, -10 mK) and CrQ-Dm_2 a smaller pulse-magnet (0.6 t, 50 T, 30 rate) sample instruments. Both are able to perform most of the investigations in horizontal scattering, but not only. For versatility and flexibility reasons, CrQ-Dm(s) have been conceived with similar configurations, having a high level of compatibility inside and outside, however exhibiting some distinct differences. The kinematic, design, simulations and precision principles applied, together with the challenging aspects and results of tests are presented.

INTRODUCTION

SwisFEL provides a very stable, compact and cost-effective driven by a low-energy and ultra-low emittance beam travelling through short-period undulators [1]. The ARAMIS [2] beamline is operating two experimental stations (ALVIRA, BERNINA). A new branch (ARAMIS 3) or, alternatively CRISTALLINA is currently under completion. Mainly, it delivers ultra-short (sub-femto) pulses (1.77-12.4 keV) X-rays to the dedicated experimental station for condensed matter physics experimental investigations. New instruments here will be focused on quantum phenomena (many body states) analysis.

However, advanced synchrotron investigations require not only improved beam characteristics and/or new modern techniques applied, but dedicated instruments adapted to the specificity of the applications.

A call for interest, including the development of two diffractometers (Dm(s)), working with specific experimental instruments to create extreme sample environmental conditions (magnetism, temperature, pressure) has been released [3]. Mainly, the intention was to effectively use them for manipulating with high precision, two heavy magnets (cryo, pulse), using ultra-fast investigative X-ray diffraction (Df) techniques. A well-known company [4] with the necessary expertise and already acquired experience [5] was in charge with the development.

The associated aspects of the inherent challenging work resulted from specifications (load, precision, environment), together with the kinematics, design solutions, fabrication and last tests for final products are presented here.

CRISTALLINA DMs

Two Dm(s) machines have to alternatively work inside of an experimental station (CRISTALLINA), using the same beamline and accommodating with the use of "diffraction-before-destruction" X-ray techniques. CrQ-Dm_1, manipulating a heavy load cryo-magnet (0.65 t, 5 T, -10 mK) and CrQ-Dm_2 a smaller (pulse) one (0.45 t, 50 T, 30 rate)), holding the sample(s) inside. Dm(s) will be permanently located nearby of the beam (preparation room), to be easily moved/removed from/in the area (and, along the beamline), connected to a fixed rails dock station.

To decrease the beam preparation waiting time, an increased compatibility between the components, and with the third one (Dm-XPP-GPS, BERNINA) have to be assured, guaranteeing easy and fast set-ups/replacements.

The first precision tests (FAT) of the developed products, being already presented [6], a short overview with updates (modeling, SAT) is done here.

Kinematics

The complexity of tasks related to the specificity of work (heavy load, high precision, strong magnetic field), together with additional requirements (efficiency) have led to considering the final kinematics Dm(s) structures (K_i, i = 1, 2), as shown in Fig. 1.

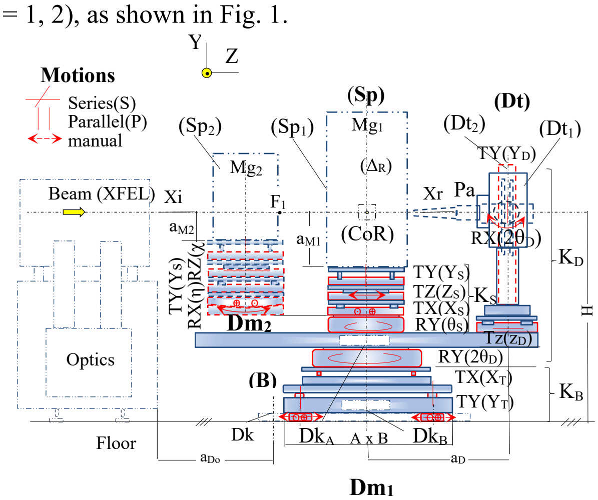

Figure 1: CrQ-Dm(s) kinematics.

($K_{1,2}$) are based on combinations of serial/parallel partial kinematic chains, corresponding to the related components of the machines (K_B, K_D, K_S). Some of them, being actuated by circles (C_i), form the system (machine) manipulator architecture (geometry). The experimental investigations are based on a corelated motions of positioning of two of them (K_S, K_D) relative to the beam. KB supports all structures

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above it, being able to perform (pre)alignment operations as well. It consists of a device table comprising an in-parallel actuated component 2-4 (TR), 2-dof (TY, RX) T-(actuated) vertical translations, R-rotation joints and a horizontal translational (TX) one. Thus, the required short (motorized) coarse (TY(Yt)=+60-20 mm, TX(Xt)=+60-30 mm, acc.=4 µm, rep.= 2 µm), in addition to (manually) fine (TY=±20 mm, acc.= 0.1 mm; RX==±20 °, acc.=0.5 °) motions, being possible to be done. Moreover, fast (re)move of entire (B) on the floor is possible with three (3) pairs of planar air pads.

(K_S) consists of two configurations corresponding to the distinct sample manipulators (S_1, S_2). (K_{S1}, K_{S2}) are based on in-series kinematic chains (devices) mechanisms. Basic devices with an actuated circle $C_{1S}(\theta)$, supporting above three translational devices with orthogonal precision motions-TX(X_S)=±25 mm, TY(Y_S)=±10 mm, TZ_S = ±25 mm (acc=1 µm, rep=<1 µm), being common. (K_{S2}) has an interposed additional device, performing two short orthogonal partial (arc) circular motions (C_{2S}, C_{3S},).

(K_D) consists of two configurations related to the detector types (D_1, D_2). (K_{D1}, K_{D2}) have in common an active circle $C_{1D}(\nu_D)$ and radial translational $(TZ)_D$ devices. (K_{D1}) holds a big detector (40kg), while (K_{D2}), a combined polar analyzer (Pa) and small detector (25 kg). (Pa) includes several actuated circles – C_{1P} ($\theta_X = \pm5°$)/C_{2P}($\delta_X = \pm10°$), C_{3P} (η_Z =+30-110°) and C_{4P} ($\chi_Y = \pm5$ °) and stage (RX).

Most important motions, together with functional and precision required parameters are included in Table 1.

Table 1: CrQ-Dm(s) Basic Motion Parameters

Type (R)	Range [°]	Acc. [arcs]	Rep. [arcs]	Circles (Ci)
RY(θ_S)	±180			$C_{1S}(\mu)$
RX(θ_S)*	±10	1	0.36	$C_{2S}(\eta)$
RZ(θ_S)*	±10			$C_{3S}(\chi)$
RY($2\theta_{D,H}$)	±180	1	0.36	$C_{1D}(\nu)$
RX($2\theta_{D,V}$)*(	±25	3	1	$C_{2D}(\delta)$

* Dm_2, (*) Optional

Reference system (O-XYZ) has right hand orthogonal axes with Z(+) along the beam and Y(+) vertically up.

Based on above structures of circle combinations, each of Dm(s) can be defined, as with Dm_1- 2C(1S+1D), and Dm_2 – 4C(3S+1(2)D) geometries.

Design

Adequate components have been chosen (or, designed), taking into account the functional specifications (load and precision) and auxiliary considerations (e.g., environment, space, safety, etc). In the design process (CREO/Pro ENGINEER), a modular approach has been adopted with the (D), (S) and (B) subsystems, as main positioning modules (Pm_i, i=1-3) built as, a combination of positioning units $(Pu)_i$ with linear/rotation active axis (A_i), Fig. 2.

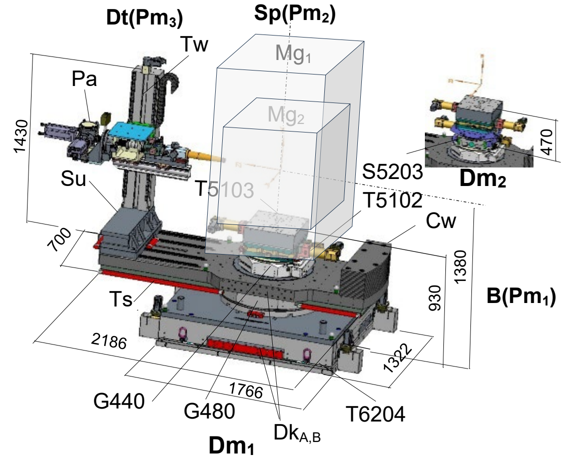

Figure 2: CrQ-Dm(s) layout (3D).

Generally, Pu_i has been chosen as commercially available (standard) devices, coming from the company store slightly modified (customized). Non-magnetic materials (stainless steel, Al, brass, etc) have been used for them to comply with the high magnetic field (10G/1m).

(Pm_1), or alignment table base (B) module mainly, includes a stiff and stabile modified device (Tb6204), which can support a heavy mass (m > 1000 kg), providing also alignment motions possibilities. It integrates two Pu_i, (i=1, 2) - table (itself), and a linear stage, respectively. (Pu_1) is based on four powerful actuated (jacks) axes (A_1-A_4), symmetrically and vertically arranged, which through the afferent gear boxes on a stiff base is driving a lighter upper plate (table). (Pu_2) being a stiff linear stage (T5101) with the actuated axis (A_5), as motor, gear box and guides (x4) embodied in the plate. The air pads which do not exceed the size of the table are hidden by shields.

$(Pm_2)_1$ corresponding to sample (S_1) module consists of a stacked combination of one Goniometer (G440) and two linear stages (T5102, T5103) positioning units (Pu_i, i=1-2). The entire module has four actuated axes (A6, A7-A8, A11). $(Pm_2)_2$ in addition has interposed a powerful stiff segment stage (S5203) with two actuated (A9, A10) axes.

The detector (D) positioning modules $(Pm_3)_{1,2}$ have been designed based on two separate devices – stiff, powerful and precision gonio (G480) having vertical actuated axis (A_{12}) and linear stage (T5101) with horizontal axis (A_{13}) for detector towers $(Tw)_{1,2}$ fixed on a common support (Su) to adapt their distance in respect with CoR axis. Both, being integrated in a stiff symmetric shape basic (arm).

$(Pm_3)_1$, corresponding to tower (Tw_1) includes a pixel detector (8M, JUNGFRAU, PSI), while $(Pm_3)_2$ configuration consisted of a combination of a polar analyzer (Pa) and small detector (1M, JUNGFRAU, PSI) subsystems held on a long linear stage (T5101) and rotation stage (G420) mounted on a vertical column (Tw_2). (Pa) includes gonio (G410A, G409), attenuator (At3000), vacuum tube (Vc) and slit (JJ) and alignment head (H1005) located on a linear (T5101) stage. ($Pm_{1,2}$) have means to be docked ($Dk_{A,B}$). An overview of the main Pm(s), Pu(s) and the corresponding actuation axis (A_i) are included, Tab. 2. As the static & dynamic behavior of a (Dm) affects the throughput (time)

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and precision (stability) of investigations, a modelling and simulation process has been performed, related to the stress and proof of strengths against load(s) and to find the smallest natural frequency mode on machines components, using Finite Element Analysis (FEA). The simulations involved the management of a large number of iterations and numerical data. As an example, in the (Dt) base case (m = 2200 kg, steel) as one of the most critical components, the first four eigenfrequencies (f_i, i=1…4) are shown in Fig. 3. The small one value is reaching the required value (f_1=40 Hz).

Table 2: CrQ - Design Components

Module	Pu	Axis	Device
Pm_1	Pu_1, Pu_2	A_1-A_4, A_5	Tb6204
Pm_2	Pu_3, Pu_4, Pu_5, Pu_6	A_6, A_7-A_8 A_9-A_{10}, A_{11}	G440, T5102.50 S5203.50, T5103.C40
Pm_3	Pu_7, Pu_8	A_{12}, A_{13}	G480, T5101.80

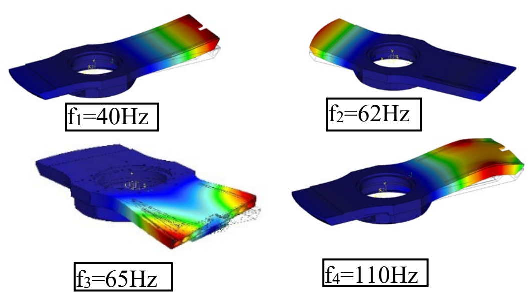

Figure 3: Natural frequency modes (Dt).

The control of the multi-axis systems (A_i, i = 13) has been generally realized, using closed loops, including servomotors (AM8121/BECKHOFF), own gears boxes (Gb2000) limit switches (24 V/NC) and absolute encoders (RESOLUTE/RENISHAW) compatibles with BiSS-C or SSI communication protocol (26-bits) performed via an Ethercat-based solution and software tool (EPICS).

To avoid unexpected collisions with adjacent optics and between own components, touch sensors (TS28/TAPE SWITCH) have been included.

Prototypes

After all, above design considerations have been taken into account and final solutions selected, the Dm(s) parts (components) have been manufactured (or, purchased). Carefully has been given in the machining and assembly processes. Then, the prototypes were tested against the specifications. The results obtained in the factory acceptance test (FAT) report included. These included functional (range, speed, etc), and precision (accuracy, repeatability, etc) parameters. In the earlier steps, individual components (stages) have been (pre)tested from accuracy point of view, as well. Global precision parameter, Sphere of Confusion $(SoC)_S$ under maximum load (M_1 = 1000 kg, M_2 = 750 kg), representing runout errors of combined (Sp) circles motions value 21/24 (load) µm was well inside of required interval. Partial values, related with one/two circles $(SoC)_{S1,2}$ were also between limits. Stepper motors have been used and own control box (SMC9300).

More recently, site acceptance tests have been performed at the customer location [7], using servomotors and dedicated controllers. The obtained values included in a report (SAT) [8], almost (re)confirmed the previously ones (factory). Calibration to compensate for the motions systematic errors(ε), for main components (re)applied. An example, for main devices (G480, G440) is given in Fig. 4. Dm_2 machine under the test is shown in Fig. 5.

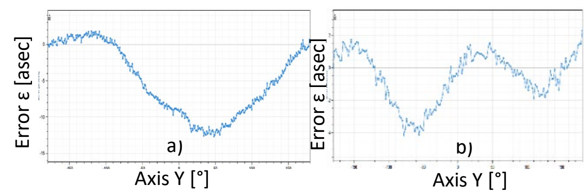

Figure 4: CrQ Positioning errors - a) ε_a = 14.5, b) ε_b = 6.5.

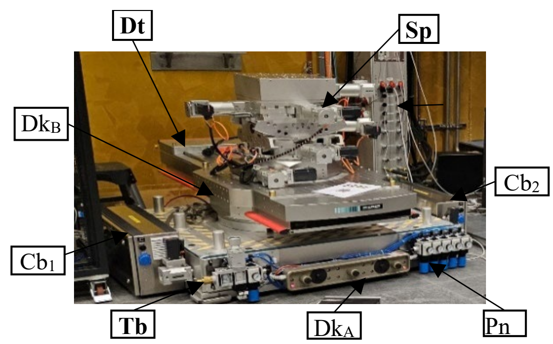

Figure 5: CrQ - Dm_2 machine (tests).

CONCLUSION

Heavy load and high precision diffractometers (CrQ-Dm(s)) for manipulating special quantum instruments with high flexibility has been developed, tested and installed, confirming the previously obtained precision motion final /partial parameters. Due to their unique features is expected that Dm(s) will fast progress the specific investigations, opening the way for another similar developments.

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